

VETRA

Product Requirements Document (PRD) Digital Livestock Healthcare Infrastructure Platform

A comprehensive, product-focused specification detailing Vetra’s market rationale, user personas, core functional modules, user journeys, MVP scope, success metrics, and competitive differentiation for product designers, business stakeholders, and competition judges.

PRODUCT
SPECIFICATION

USER PERSONAS &
JOURNEYS

MVP
REQUIREMENTS

BUSINESS MODEL &
ROADMAP

DOCUMENT ID & TYPE

VETRA-PRD-2026 // Master Product Specification

PRODUCT CATEGORY

Digital Healthcare Infrastructure for Livestock

TARGET AUDIENCE

Founding Team, Designers, Judges & Stakeholders

DOCUMENT STATUS

Active / Competition & Product Reference

SECTION 1

Product Overview, Vision & Table of Contents

What Vetra is, the fundamental vision, and the complete structure of this specification.

WHAT IS VETRA?

Vetra is a connected digital livestock healthcare platform that creates a persistent health identity for every animal, maintains immutable veterinary medical history, connects farmers with licensed veterinarians, and provides context-aware assistive AI-powered health screening.

Product Vision

Transform livestock healthcare from fragmented, reactive crisis treatment into **continuous, data-driven animal healthcare** across agricultural economies.

Product Mission

Ensure every livestock animal has an accessible digital health history, early symptom awareness, and structured veterinary clinical support.

Product Category

Digital Healthcare Infrastructure for **Livestock** — a unified operating layer connecting animals, farmers, veterinarians, and institutions.

Document Structure & Navigation

- **Page 3:** Section 2 — Problem Statement & Real-World Breakdowns
- **Page 4:** Section 3 — Target Users & Stakeholder Personas
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Product Summary Anchor: "Vetra is not an isolated AI toy or a generic ledger. It is the persistent digital infrastructure that organizes the complete clinical lifecycle of livestock."

SECTION 2

Problem Statement: The 5 Broken Healthcare Gaps

A detailed breakdown of industry breakdowns, their real-world impact, and how Vetra solves each.

1. Lost & Fragmented Medical History

Problem: Animals have no permanent health records. Vaccinations, past treatments, and chronic illnesses are stored on paper slips or remembered verbally.

Impact: Visiting veterinarians start from zero, resulting in trial-and-error treatments, repeated misdiagnoses, and unmonitored antibiotic overuse.

How Vetra Solves It: Creates an immutable digital **Animal Passport & EVMR** tied to the ear tag that permanently records every clinical event.

2. Severe Veterinary Accessibility Friction

Problem: Farmers struggle to find available, qualified veterinarians quickly, relying on informal phone calls or traveling miles to distant clinics.

Impact: Critical treatment delays of 24–72 hours, causing easily treatable infections to progress into fatal systemic conditions.

How Vetra Solves It: A structured **1–Tap Veterinary Booking Machine** matching farmers with nearby registered doctors based on availability.

3. Late Disease Detection & Triage Vacuum

Problem: Early symptoms (minor appetite drop, subtle lethargy, mild discharge) are ignored because farmers hesitate to pay for a full vet visit.

Impact: High livestock mortality, drop in daily milk yield, and missed opportunities for low-cost supportive first-aid.

How Vetra Solves It: **Context-Aware AI Screening** (Visual Scanner & Advisor) providing early risk assessment and prompt vet referral.

4. Lack of Healthcare Coordination

Problem: Farmers, private practitioners, government vets, and dairy cooperatives operate in complete isolation without shared data.

Impact: Inefficient veterinary travel routes, unverified insurance claims, and fragmented rural healthcare delivery.

How Vetra Solves It: A **unified dual-role platform** where pre-visit data, clinical charts, and follow-ups synchronize in one ecosystem.

5. Zero Livestock Health Intelligence & Blind Outbreak Spread

The Problem: Contagious livestock epidemics (Lumpy Skin Disease, FMD) propagate silently because paper records remain trapped in individual barns.	The Impact: Millions of cattle infected, trade bans, and catastrophic economic destruction for rural farming communities.	Vetra's Solution: Aggregates verified EVMR clinical diagnoses geographically to deliver real-time spatial disease intelligence and early outbreak warnings.
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Product Rationale: "Vetra addresses the full continuum of livestock care: from lost records and triage delays to clinical coordination and regional epidemic containment."

SECTION 3

Target Users & Stakeholder Personas

Detailed profiles, goals, pain points, and specific product requirements for each user group.

 **Persona 1: Livestock Farmer / Smallholder Owner**

PRIMARY B2C / USER

Core Goals:

- Protect animal health & life
- Maintain daily milk income
- Fast access to qualified doctors

Daily Pain Points:

- No persistent medical records
- Uncertainty during early illness
- Struggle to contact available vets

Needs from Vetra:

- Digital Animal Passport
- Context-Aware AI Advisor triage
- 1-Tap Vet Consultation booking

 **Persona 2: Licensed Field Veterinarian**

PRIMARY PRACTITIONER

Core Goals:

- Deliver superior clinical care
- Manage daily farm rounds
- Build trusted client relationships

Daily Pain Points:

- No pre-visit animal history
- 1.5+ hours of daily paperwork
- Unorganized WhatsApp messages

Needs from Vetra:

- Immutable EVMR creation (<45s)
- Complete patient clinical timeline
- Structured appointment roster

 **Persona 3: Dairy Cooperatives, Insurers & Institutions**

ENTERPRISE STAKEHOLDER

Core Goals:

- Secure raw milk supply chain
- Prevent contagious herd epidemics
- Verify livestock insurance claims

Daily Pain Points:

- Zero real-time disease visibility
- High fraud in insurance settlements
- Unmonitored antibiotic residues

Needs from Vetra:

- Verified medical history feeds
- Regional disease intelligence maps
- Biosecurity compliance reports

Persona Alignment: "Vetra creates a multi-stakeholder ecosystem where solving the farmer's immediate pain creates the data foundation that empowers veterinarians and protects dairy cooperatives."

SECTION 4

Product Vision, Core Tenets & Clinical Boundaries

The foundational product philosophy and non-negotiable architectural boundaries governing Vetra.

1. Animal-Centric Healthcare

The individual animal is the permanent unit of healthcare. Every clinical chart, AI scan, vaccination, and owner transfer permanently anchors to the animal's lifetime passport.

2. Human-in-the-Loop

Licensed veterinarians remain the ultimate clinical authority. Technology provides data and triage, but clinical diagnosis, examinations, and drug prescribing remain in human hands.

3. Assistive, Not Autonomous AI

Artificial intelligence acts exclusively as an assistive screening bridge to catch symptoms early and guide supportive care—never as an autonomous diagnostic agent.

NON-NEGOTIABLE PRODUCT RULE: WHAT VETRA IS NOT

✗ NOT an AI Doctor

Vetra never claims to replace physical veterinary auscultation, palpation, or clinical judgment.

✗ NO Autonomous Prescriptions

AI models are strictly forbidden from generating pharmaceutical drug names or chemical dosages.

✗ NOT a Commodity Marketplace

Vetra is pure healthcare infrastructure, not a livestock brokerage or classifieds app.

THE TRUE NATURE OF THE VETRA PLATFORM

"Vetra is the digital operating infrastructure connecting animals, farmers, licensed veterinarians, and public health intelligence into one continuous, self-reinforcing circle of care."

Product Guideline: "In all UI copy, marketing, and investor materials: Always emphasize human clinical authority and assistive decision support."

SECTION 5.1 & 5.2

Core Features: Animal Passport & EVMR System

Detailed product specifications, problems solved, user workflows, and expected outcomes.

FEATURE 1: Animal Passport (Digital Identity & Health Profile)

CORE FOUNDATION

Purpose: Provide every livestock animal with a permanent, tamper-proof digital health identity linked to its physical ear tag or QR code.

User Problem Solved: Eliminates lost health records, unverified ownership claims, and the absence of clinical memory during veterinary visits.

User Workflow: Farmer registers animal in <60s with tag number, species, breed, and gender → Unique QR Passport generated → Passport displays lifetime timeline.

Expected Outcome: 100% of animal medical events, vitals, and vaccinations attach to one accessible profile.

FEATURE 2: Electronic Veterinary Medical Record (EVMR)

CLINICAL CORE

Purpose: Standardize clinical documentation through immutable digital charts created exclusively by licensed veterinarians.

User Problem Solved: Replaces messy paper slips, illegible prescriptions, and unverified medical history with structured legal health records.

User Workflow: Vet completes clinical examination → Enters diagnosis, symptoms, treatments, vitals, and prescriptions in <45s → Submits immutable record.

Expected Outcome: Permanent, unalterable clinical history attached directly to the animal's lifetime passport.

Structured Clinical Data Model of EVMR

- Confirmed Diagnosis
- Primary Symptoms

- Clinical Treatment
- Prescriptions Issued

- Body Temp (°F/°C)
- Live Weight (kg)

- Follow-up Date
- Clinical Notes

Product Standard: "The Animal Passport is the container; the EVMR is the clinical truth. Together, they create unbreakable medical continuity."

SECTION 5.3 & 5.4

Core Features: Veterinary Booking & AI Scanner

Automating practitioner dispatch and assistive visual disease screening in the field.

FEATURE 3: Veterinary Booking & Dispatch System

Purpose: Connect farmers with qualified, registered veterinarians through a reliable, deterministic appointment state machine.

User Problem Solved: Eliminates informal phone coordination friction, missed appointments, and unverified practitioner credentials.

User Workflow: Farmer selects sick animal & visit type → Requests appointment → Vet receives pre-visit context & confirms → Consultation completed.

Expected Outcome: Average dispatch coordination time reduced from 24+ hours to under 30 minutes.

OPERATIONS

FEATURE 4: AI Disease Scanner (Assistive Visual Vision)

Purpose: Provide farmers with an instant visual screening tool to analyze visible lesions, skin conditions, and ocular/nasal discharges.

User Problem Solved: Removes guesswork when physical symptoms appear, catches contagious signs early, and encourages timely vet care.

User Workflow: Farmer captures/uploads symptom photo → AI analyzes morphological patterns → Delivers preliminary assessment + confidence + vet referral.

Expected Outcome: Immediate first-line triage guidance without false medical certainty.

VISION AI

MANDATORY SCANNER OUTPUT ELEMENTS (NEVER CALL IT DIAGNOSIS)

1. **Suspected Condition:** Categorized preliminary pattern match.

2. **AI Confidence Score:** Model probability clearly framed.

3. **Veterinary Review CTA:** Direct 1-tap booking handoff.

Product Requirement: "Every visual scan result MUST display the mandatory veterinary disclaimer and provide a 1-tap route to book a professional consultation."

SECTION 5.5

Core Features: Context-Aware AI Advisor

Conversational clinical triage anchored to the individual animal's historical health context.

FEATURE 5: AI Veterinary Advisor (Context-Aware Interactive Triage)

TRiage AI

Purpose: Guide animal owners through structured conversational health screening when an animal is unwell, asking targeted follow-up questions and producing a preliminary risk assessment.

HOW CONTEXT AWARENESS WORKS (THE KEY PRODUCT DIFFERENTIATOR)

- **Species & Breed:** Cattle vs. Buffalo vs. Goat biological norms.

- **Historical EVMRs:** Past confirmed illnesses & treatments.

- **Recent Scans:** Integrates visual telemetry into chat context.

Interactive Triage Mechanics

- **Natural Language Input:** Farmer describes symptoms in simple vernacular text or voice.
- **Dynamic Suggestion Chips:** Generates 1-tap responses (e.g., *"Drinking normally"*, *"No coughing"*) for fast field interaction.
- **Strict Fact Preservation:** Preserves user-reported facts accurately and never fabricates denied symptoms.

Structured Assessment Card

- **Risk Level Classification:** Categorized as LOW, MODERATE, HIGH, or CRITICAL.
- **Supportive Care Guidance:** Non-prescriptive first-aid (hydration, isolation, temperature monitoring).
- **1-Tap Vet Consultation Handoff:** Pre-fills doctor booking form with complete conversation history.

Product Summary: "Vetra's AI Advisor reads the animal's health chart before asking its first question, delivering grounded preliminary guidance and seamless doctor referrals."

SECTION 6

End-to-End User Journeys: Farmer & Veterinarian

Visual step-by-step product interaction workflows from symptom onset to permanent record closure.

Journey 1: Farmer with a Sick Animal (e.g., Ramesh & Gauri)

1. Symptom
Animal unwell

→

2. Passport
Opens profile

→

3. AI Advisor
Context triage

→

4. Assessment
Risk & care advice

→

5. Book Vet
1-Tap dispatch

→

6. EVMR
Record locked

Farmer Value: Reduces anxiety in the barn, provides safe immediate supportive guidance, and connects with a doctor in under 2 minutes.

Journey 2: Field Veterinarian Clinical Practice Workflow (e.g., Dr. Roy)

1. Alert
Request received

→

2. Context
Reviews history

→

3. Confirm
Accepts visit

→

4. On-Farm
Examines animal

→

5. EVMR Form
Inputs clinical chart

→

6. History
Permanently saved

Veterinarian Value: Eliminates blind visits, streamlines daily travel routes, and cuts paperwork time from hours to under 45 seconds per animal.

Product Symphony: "The farmer journey and veterinarian journey interlock seamlessly. The farmer's triage initiates the visit; the veterinarian's examination closes the clinical loop."

SECTION 7.1

Product Requirements Matrix: Identity, EVMR & Booking

Formal functional requirements, traceability IDs, descriptions, and prioritization.

FEATURE ID	MODULE	FUNCTIONAL REQUIREMENT	PRIORITY
AP-001	Animal Passport	Every livestock animal must have a globally unique digital identity linked to its ear tag number.	MUST HAVE
AP-002	Animal Passport	System must generate a scannable QR code for instant field retrieval of the Animal Passport.	MUST HAVE
AP-003	Animal Passport	Passport must render a chronological timeline of all completed medical records and scans.	MUST HAVE
MR-001	EVMR System	Only verified, licensed veterinarians can create an official EVMR clinical record.	MUST HAVE
MR-002	EVMR System	EVMR records must be immutable once created—no modification or deletion allowed.	MUST HAVE
MR-003	EVMR System	EVMR must capture confirmed diagnosis, treatment, prescriptions, temperature, and follow-up.	MUST HAVE
BK-001	Vet Booking	Farmers must be able to search and select registered veterinarians based on location & specialty.	MUST HAVE
BK-002	Vet Booking	Appointments must follow a formal state machine: PENDING → CONFIRMED → COMPLETED.	MUST HAVE
BK-003	Vet Booking	System must pre-fill the booking request with the sick animal's telemetry and symptoms.	SHOULD HAVE

Priority Definitions		
MUST HAVE (P0): Essential core functionality required for product launch.	SHOULD HAVE (P1): High-value workflow enhancements for optimal experience.	COULD HAVE (P2): Extended features planned for subsequent version releases.

SECTION 7.2

Product Requirements Matrix: AI, Triage & Profiles

Requirements governing AI screening, conversational triage, user profiles, and security.

Feature ID	Module	Functional Requirement	Priority
AI-001	AI Scanner	Camera module must accept photos and return preliminary visual assessments in <5 seconds.	MUST HAVE
AI-002	AI Scanner	All visual scan outputs must explicitly display AI Confidence and veterinary review disclaimers.	MUST HAVE
AI-003	AI Advisor	Advisor session must ingest animal species, age, and historical EVMRs as clinical context.	MUST HAVE
AI-004	AI Advisor	Advisor must enforce strict fact preservation and never fabricate user-denied symptoms.	MUST HAVE
AI-005	AI Advisor	System must trigger prominent emergency alert banners for critical risk symptom inputs.	MUST HAVE
PR-001	User Profiles	System must enforce strict role separation between Farmer and Veterinarian accounts.	MUST HAVE
PR-002	User Profiles	Veterinarian profiles must capture clinical credentials, qualifications, and clinic location.	MUST HAVE
NT-001	Notifications	System must alert farmers and veterinarians upon appointment confirmation and status updates.	SHOULD HAVE
OF-001	Offline Mode	Mobile app must cache passports and recent EVMRs locally for offline viewing in remote barns.	SHOULD HAVE

Quality Standard: "All PO (Must Have) requirements are fully implemented, tested, and validated against the live staging cloud environment."

SECTION 8

MVP Scope vs. Future Capabilities

Clear boundary separation between core launch capabilities and long-term roadmap features.

✔ Core MVP Scope (Production Ready)

CURRENT BUILD

1. Digital Animal Passport:

Full animal registration, ear tag indexing, telemetry, and lifetime timeline.

2. Immutable EVMR System:

Standardized veterinary charts, diagnosis, treatments, and prescriptions.

3. Role-Based Auth & Profiles:

Dual-role Farmer & Veterinarian onboarding, security isolation.

4. Appointment State Machine:

Deterministic booking, confirmation, and visit completion workflow.

5. AI Disease Scanner:

Visual lesion screening, AI confidence scores, and supportive care advice.

6. AI Veterinary Advisor:

Context-aware conversational triage, fact preservation, and doctor handoff.

🚀 Future Capabilities (Roadmap)

PLANNED EXPANSION

1. PostGIS Outbreak GIS Mapping:

Real-time spatial disease cluster heatmaps and quarantine zones.

2. Livestock Insurance Integration:

Automated claim verification APIs for rural insurers.

3. Government Surveillance Portals:

Regional public health dashboards for state animal husbandry.

4. Enterprise Herd Analytics:

Bulk milk yield correlation and vaccination compliance reports.

5. IoT Wearable Sensor Hooks:

Bluetooth collar telemetry (rumination, estrus, body temp).

EXPLICIT NON-GOALS FOR VETRA PLATFORM

• No direct livestock e-commerce buying/selling transactions.

• No autonomous pharmaceutical prescription generation.

• No replacement of official government regulatory census.

Product Discipline:

"Maintaining strict scope boundaries ensures rock-solid clinical stability for our core MVP while laying clean architectural hooks for future scale."

VETRA PRODUCT REQUIREMENTS DOCUMENT // CONFIDENTIAL

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SECTIONS 9 & 10

Success Metrics & Commercial Business Model

Measurable performance indicators and viable commercial pathways for agricultural markets.

User Adoption Metrics

- Total Registered Animals
- Monthly Active Farmers (MAF)
- Verified Veterinarian Retention
- QR Passport Scan Frequency

Clinical Healthcare Metrics

- EVMR Generation Rate (>95%)
- Reduction in Care Delays (<2h)
- Completed Consultations
- Early Triage Escalation Rate

Platform Operations Metrics

- Appointment Completion Rate
- AI Advisor Triage Conversion
- Offline Sync Success Rate (99.9%)
- Zero Data Loss SLA

COMMERCIAL MONETIZATION PATHWAYS (HIGH-LEVEL FRAMEWORK)

1. B2B Veterinary SaaS:
Clinic practice subscriptions for multi-doctor scheduling, billing, and EVMR tools.

2. B2B2C Dairy Co-ops:
Enterprise contracts with dairy unions to deploy passports and biosecurity across farmer bases.

3. Institutional APIs:
Verification APIs for livestock insurance underwriting and government epidemic tracking.

Financial Integrity Note: "Vetra focuses on validated clinical and market utility rather than inventing speculative revenue projections."

SECTIONS 11 & 12

Competitive Positioning & Multi-Phase Roadmap

How Vetra wins against legacy alternatives and our phased strategic horizon.

CATEGORY	TRADITIONAL PRACTICE	GENERIC AI / CHATBOTS	FARM LEDGER APPS	VETRA PLATFORM
Medical Memory	Lost paper slips	Zero historical context	Non-clinical milk logs	Immutable Digital Passport (EVMR)
Clinical Care	Informal phone calls	None (stateless text)	None	1-Tap Veterinary Dispatch
Triage AI	None	Generic hallucinations	None	Context-Aware Assistive Triage
Surveillance	Zero regional visibility	None	Isolated on single farm	Regional Outbreak Intelligence

STRATEGIC MULTI-PHASE ROADMAP

Phase 1: Identity & Clinical Core (Live)

Animal Passport, EVMR creation, vet booking state machine, cloud & mobile clients.

Phase 2: AI Screening & Intelligence (Live)

AI Disease Scanner, context-aware AI Advisor, safety guardrails, offline caching.

Phase 3: National Scale Ecosystem

Spatial GIS maps, cooperative bulk portals, livestock insurance APIs, national registries.

Strategic Differentiator: "Point solutions solve isolated tasks. Vetra builds the complete, connected digital infrastructure for livestock healthcare."

SECTION 13

Product Risks, Clinical Safety & Ethical AI

Architectural safeguards, data privacy, and legal frameworks ensuring zero patient harm.

1. AI Limitations & Boundary Enforcement

Vetra AI models are engineered with hard systemic boundaries: they are **strictly non-prescriptive**, mathematically incapable of generating drug names or antibiotic dosages, and always label statistical confidence transparently.

2. Veterinary Authority & Responsibility

The licensed veterinarian retains exclusive legal authority over clinical diagnosis, physical examinations, surgical interventions, and pharmaceutical prescriptions within the EVMR module.

3. Data Privacy & Farmer Ownership

Livestock records are private to the registered farmer and treating veterinarian. Aggregated epidemiological disease intelligence is completely anonymized to protect farm commercial privacy.

4. Conservative Emergency Escalation

In cases of life-threatening distress or ambiguous symptoms, Vetra prioritizes **URGENT VETERINARY REVIEW** over false reassurance, ensuring critical animals receive immediate doctor dispatch.

MANDATORY CLINICAL SAFETY STATEMENT

"Vetra's AI screening tools are assistive decision-support aids designed to catch symptoms early. They do not constitute a veterinary diagnosis and never replace the clinical expertise of a licensed veterinary practitioner."

Ethics Anchor: "By designing for safety from day one, Vetra establishes trust with farmers, clinical practitioners, dairy unions, and veterinary boards."

SECTION 14

Vetra Product Vision & Closing Manifesto

The enduring mission and commitment that defines the Vetra platform.

*"Every livestock animal deserves a lifelong health record,
every farmer deserves accessible veterinary support,
and every veterinarian deserves better information to provide care."*

What Vetra Accomplishes

Vetra bridges the critical divide in livestock healthcare. By giving animals a permanent digital health identity and connecting farmers, doctors, and intelligence, we ensure that no animal's medical history is lost and no treatable illness is ignored.

The Path Ahead

As Vetra scales from individual farms to national dairy networks, it becomes the trusted digital infrastructure that safeguards livestock productivity, food security, and rural prosperity across emerging agricultural economies.

VETRA PRODUCT SPECIFICATION // END OF DOCUMENT
DIGITAL LIVESTOCK HEALTHCARE INFRASTRUCTURE PLATFORM

VETRA PLATFORM // PRODUCT MANAGEMENT & DESIGN // APPROVED SPECIFICATION // 2026